RACE
Regular Analysis Through Continuous Exercise

Area Under The Curve: RACE-01

Subjective Type Questions :

1. Find the area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = -\frac{1}{2}$ and $x = 1$.

2. Find the area bounded by $x = 2, x = 5, y = x^2, y = 0$.

3. Find the area enclosed between $y = \cos x$ & x-axis as x varies from $\frac{-\pi}{2}$ to $\frac{3\pi}{2}$

4. Find area enclosed between $y = \ell nx$, x-axis and the two ordinates $x = \frac{1}{e^2}$ and $x = e^2$.

5. Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes

6. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ell n\, x$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1, C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of $'a'$.

7. Find the positive value of $'a'$ for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices $(0, 0), (a, 0), (0, a^2 + 1)$ and $(a, a^2 + 1)$.

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8. Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$

9. Find the values of m $(m > 0)$ for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is, (i) 9/2 square units & (ii) minimum. Also find the minimum area.

10. If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is least. Find k and also the least area.

Answer Key

1. $(\frac{3}{8}$ sq. units) **2.** $(\frac{117}{3}$ sq. units) **3.** (4 sq. units)

4. $(2 + e^2 - \frac{3}{e^2}$ sq. units) **5.** $(\frac{56}{3}$ sq. units) **6.** (e)

7. $(\sqrt{3})$ **8.** $(\frac{4 - 3\ell n3}{2})$

9. ((i) m = 1, (ii) m = ∞ ; A_{min} = 4/3) **10.** $(k = 2, A_{least} = \frac{32}{3})$

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Area Under The Curve: RACE-02

Subjective Type Questions :

1. Find the area bounded by the curves $x = y^2$ and $x = 3 - 2y^2$.

2. Find the area of the region bounded by the curves $x = \frac{1}{2}, x = 2, y = \log x$ and $y = 2^x$.

3. Find the area of the region $\{(x, y) : 0 \le y \le x^2 + 1, 0 \le y \le x + 1, 0 \le x \le 2\}$.

4. A figure is bounded by the curves $y = \left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, $y = 0, x = 2$ & $x = 4$. At what angles to the positive x-axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same area.

5. The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

6. Find the area enclosed between $y = \sin x$; $y = \cos x$ and y-axis in the 1st quadrant.

7. Find the area bounded by the inverse of bijective function $f(x) = 4x^3 + 6x$, the x-axis and the ordinates $x = 0$ & $x = 44$.

8. Find area bounded by $y = f(x)$ & $y = f^{-1}(x), x = 1$ & $x = e$ where $f(x) = e^x$.

9. Let $f(x) = x^3 + 3x + 2$ and $g(x)$ is the inverse of it. Find the area bounded by $g(x)$, the x-axis and the ordinate at $x = -2$ and $x = 6$.

10. Find area bounded by $y = f^{-1}(x), x = 10, x = 4$ and x-axis given that area bounded by $y = f(x)$, $x = 2, x = 6$ and x-axis is 30 sq. units, where $f(2) = 4$ and $f(6) = 10$. (given $f(x)$ is an invertible function)

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Answer Key

1. (4 sq. units)

2. $\left(=\frac{4-\sqrt{2}}{\log_e 2} - \frac{5}{2}\log_e 2 + \frac{3}{2}\text{ sq. units}\right)$

3. (23/6 sq. units)

4. $\left(\pi - \tan^{-1}\frac{2\sqrt{2}}{3\pi}; \pi - \tan^{-1}\frac{4\sqrt{2}}{3\pi}\right)$

5. $\left(\frac{3\pi + 2}{\pi - 2}\right)$

6. $(\sqrt{2} - 1)$

7. (60 sq. units)

8. $(e^e - e - 1)$

9. (9/2)

10. (22)

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